WORKED SOLUTION

STEP I 1993, Question 14

Probability and Statistics • Exponential Waiting Times

STEP I 1993, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q14

The key idea

What matters is not Mr Toad's speed but the *time* T he takes to cover the last kilometre, and $T = 1/V$. The density of V looks awkward, but its cumulative distribution function is very simple, and converting it into a statement about T shows that T is exponential with parameter α. After that, each green period is a difference of two exponentials, and adding up all the green periods gives a geometric series.

Solution

The distribution of the arrival time

Let V be the speed in km per minute, and let $T = 1/V$ be the time in minutes, measured from the moment the lights turn green, at which he reaches the lights. Spotting that

$$\frac{\mathrm{d}}{\mathrm{d}v}\,\mathrm{e}^{-\alpha/v} = \alpha v^{-2}\mathrm{e}^{-\alpha/v},$$

and that $\mathrm{e}^{-\alpha/v} \to 0$ as $v \to 0^+$, the distribution function of V is

$$\mathrm{P}(V \le v) = \int_0^v \alpha u^{-2}\mathrm{e}^{-\alpha/u}\,\mathrm{d}u = \mathrm{e}^{-\alpha/v} \qquad (v > 0).$$

(It tends to 1 as $v \to \infty$, so this is a genuine density.) Since $V > 0$, for $t > 0$ we have $T > t \iff 1/V > t \iff V < 1/t$, so

$$\mathrm{P}(T > t) = \mathrm{P}\left(V < \tfrac{1}{t}\right) = \mathrm{e}^{-\alpha t}.$$

So T has the exponential distribution with parameter α.

Ask yourself Why work with T rather than V?

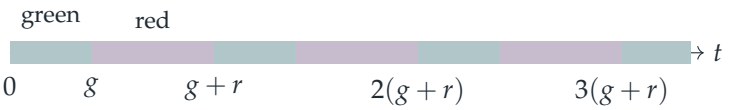
The lights are described in terms of time, so every event we care about is an interval of values of T. Translating the density of V into the density of T by hand means a change of variable; going through $\mathrm{P}(T > t)$ avoids that and gives the answer in one line.

One green period

At time 0 the lights turn green. They are green on $[0, g)$, red on $[g, g+r)$, green on $[g+r, (g+r)+g)$, and so on. The $(n+1)$th green period is $n(g+r) \le t < n(g+r)+g$. Because T is continuous, whether the endpoints are included makes no difference, and

$$\begin{aligned} \mathrm{P}(n(g+r) < T < n(g+r)+g) &= \mathrm{P}(T > n(g+r)) - \mathrm{P}(T > n(g+r)+g) \\ &= \mathrm{e}^{-\alpha n(g+r)} - \mathrm{e}^{-\alpha(n(g+r)+g)}, \end{aligned}$$

as required. (The printed paper puts the closing bracket in the wrong place in the second exponent; the exponent should be $-\alpha(n(g+r)+g)$, as here.) The working holds equally for $n = 0$, which is the first green period, before the lights have ever turned red.



The probability of a green light

He passes on green exactly when T lies in one of these green periods. They are disjoint, so we add their probabilities, including $n = 0$:

$$\begin{aligned} \mathrm{P}(\alpha) &= \sum_{n=0}^{\infty} \left(\mathrm{e}^{-\alpha n(g+r)} - \mathrm{e}^{-\alpha n(g+r)} \mathrm{e}^{-\alpha g} \right) = (1 - \mathrm{e}^{-\alpha g}) \sum_{n=0}^{\infty} (\mathrm{e}^{-\alpha(g+r)})^n \\ &= \frac{1 - \mathrm{e}^{-\alpha g}}{1 - \mathrm{e}^{-\alpha(g+r)}}, \end{aligned}$$

summing the geometric series, which converges because $0 < \mathrm{e}^{-\alpha(g+r)} < 1$ (since $\alpha > 0$ and $g + r > 0$).

The limits

As $\alpha \to \infty$, both $\mathrm{e}^{-\alpha g} \to 0$ and $\mathrm{e}^{-\alpha(g+r)} \to 0$ (as $g > 0$, $g + r > 0$), so

$$\mathrm{P}(\alpha) \to \frac{1-0}{1-0} = 1.$$

This makes sense: a large α means a very fast Mr Toad, who almost always gets through before the first green period ends.

As $\alpha \to 0$ both numerator and denominator tend to 0, so we need more care. Write $1 - \mathrm{e}^{-x} = \mathrm{e}^{-x}(\mathrm{e}^{x} - 1)$ in each:

$$\mathrm{P}(\alpha) = \frac{\mathrm{e}^{-\alpha g}\,(\mathrm{e}^{\alpha g} - 1)}{\mathrm{e}^{-\alpha(g+r)}\,(\mathrm{e}^{\alpha(g+r)} - 1)} = \mathrm{e}^{\alpha r} \cdot \frac{\dfrac{\mathrm{e}^{\alpha g} - 1}{\alpha g}}{\dfrac{\mathrm{e}^{\alpha(g+r)} - 1}{\alpha(g+r)}} \cdot \frac{g}{g+r},$$

where we divided top and bottom by α and inserted the factors g and $g+r$. As $\alpha \to 0$, $e^{\alpha r} \to 1$, and both $\alpha g \to 0$ and $\alpha(g+r) \to 0$, so by $(e^x - 1)/x \to 1$ the two fractions each tend to 1. Therefore

$$P(\alpha) \to \frac{g}{r+g} \qquad \text{as } \alpha \to 0.$$

Here Mr Toad is very slow and his arrival time is spread over a great many cycles, so the chance of green is just the proportion of each cycle that is green.

Checks

- With $\alpha = 0.7$, $g = 1$, $r = 2$ the formula gives $P(\alpha) \approx 0.5737$. A simulation of 4×10^5 drives, sampling V by inverting $e^{-\alpha/v}$, gave 0.5744, and the individual green periods $n = 0, 1, 2$ matched too.
- With $g = 1$, $r = 2$: $P(10^{-4}) \approx 0.33337$, close to $\frac{1}{3}$, and $P(10) \approx 0.99995$.
- If $r = 0$ the light is always green and the formula gives $P(\alpha) = 1$ for every α, as it should.

Common slips

- Leaving out $n = 0$. The first green period, before any red, carries the most probability.
- Copying the printed exponent $-\alpha(n(g+r)) + g$, which is a misprint for $-\alpha\big(n(g+r)+g\big)$.
- Treating V itself as the arrival time, or trying to find $E(T)$. In fact $E(V)$ is infinite here, but nothing in the question needs it.

Beyond STEP

Turning a distribution for V into one for $T = g(V)$ through the distribution function is the standard method for transforming random variables. Here V has an inverse-exponential distribution. The limit $\alpha \to 0$ is a small example of a general idea: a random time spread over many periods lands in each phase in proportion to that phase's length, which is the start of renewal theory. Grimmett and Welsh, *Probability: An Introduction*, covers both transformations of random variables and the exponential distribution.

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